

Maintenance, Care, and Troubleshooting

System Overview:

The system consists of four main technical subsystems: the Raspberry Pi controller, the DFPlayer Mini audio module, the refrigerator habitat, and the UV lighting system. During normal operation, the Raspberry Pi coordinates the exterior LEDs, interior lights, audio playback, narration cues, and UV exposure timing.

At the press of the start button, the controller begins the programmed sequence. The exterior lights change from green to red, classical music stops, and narration begins. After the first narration cue, the interior lights turn off, and the UV light inside the refrigerator turns on. After the second cue, all lights turn off for 30 seconds. The interior lights then return, and the exterior lights change from red back to green, indicating that the cycle is complete.

A diagram is included at the end of this document.

Safety Notes

- Never look directly at the UV light while it is powered.
- Disconnect power before servicing wiring, LEDs, transistors, the audio module, or the Raspberry Pi.
- Do not run the UV cycle with the refrigerator open unless the UV light is disabled.
- Verify UV exposure time after any software, wiring, or hardware change.
- Do not operate the system if the refrigerator temperature, humidity, or ventilation conditions are outside the safe range for the millipedes.
- Keep moisture away from the Raspberry Pi, DFPlayer Mini, and exposed wiring.

Specifics of the system are as follows:

- The Controller should not exceed 70°C, doing so would put the controller at risk
 - If overheating occurs, power down the system and allow it to cool. Additionally ensure there is no external heat from the refrigerator or lighting
 - The system, while autonomous once started, can be easily modified via external computer using command ``ssh user@ip`` at the terminal
- To ensure the system is properly functioning, all lights should be checked, and the UV should be timed to ensure there is no overexposure to the millipedes

Troubleshooting Guide

Problem	Likely Cause	Check
System does not start after button press	Button, wiring, controller script, Raspberry Pi power	Check Pi power, button connection, and whether <code>pi4.py</code> is running
Exterior LED does not change color	LED, transistor, resistor, 12V supply, GPIO signal	Test LED, then transistor, then GPIO output
One LED is out	Failed LED or local wiring	Replace/test LED first
All LEDs in a group are out	Power supply, common ground, transistor, controller signal	Check 12V line and GPIO control
Audio does not play	Speaker, DFPlayer Mini, microSD card, audio file issue	Check DFPlayer LED, speaker wiring, SD card contents
DFPlayer LED is off	No power, failed module, SD card issue	Check module power and microSD
UV light does not turn on	UV bulb/LED, transistor/relay, power supply, GPIO signal	Test UV circuit separately with caution
UV light stays on too long	Software timing issue, stuck transistor/relay, script crash	Cut power immediately and inspect code/hardware
Raspberry Pi cannot be reached over SSH	Network issue, wrong IP, Pi not powered, SSH disabled	Check IP address, power, and network connection
Refrigerator not cooling	Refrigerator power, thermostat, compressor issue	Confirm power and temperature setting

Additional Notes:

Files on the DFPlayer should be named 001.mp3/wav and 002.mp3/wav for the idle and narration tracks respectively.

Keep a usb with each file and script separately in case of component failure.

Replacement Part List:

- Raspberry Pi 4 (controller)
 - If this is replaced, make sure the script [pi4.py](#) is transferred
- DFPlayer mini (audio module)
- Gikfun Speaker (4Ohm 3W)
- 2N2222 Transistor
- Red/Green KEING Exterior LED (12V)
- MicroSD card
- 1k Resistor

